

AMG5122 INDUSTRIAL DUAL CHANNEL 3G-SDI WITH SERIAL DATA



Industrial Fiber Optic Solutions

The AMG5122 series are designed for transmission of two channels of SD/HD/3G-SDI video along with one RS422 serial data channel. Available in a compact aluminium housing allows for full integration within custom video and data systems.



 3G SDI x2	 Data 1x RS422	 Outputs x3	 Fiber Singlemode	 Temp -40°C~+75°C	 Mounting DIN/Surface	 Protection IP40	 NDAA/TAA Compliant
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[AMG5122 Series]

/ OVERVIEW

The AMG5122 high definition fiber video converter is an ideal solution for the transmission of broadcast or surveillance applications. It supports 2 channels of SD, HD or 3G-SDI video, supporting image transmission of SDTV (143-540Mbps) or HDTV (1.485Gbps & 3Gbps, 1080P60).

The units provide full equalizing and re-clocking of the SDI video in order to reconstruct the signal for consistent quality. Electronic and optical adjustments are never required.

An RS422 serial data channel is supported along with 3 open collector alarm outputs that provide indication of the status of the unit.

The integrated fully enclosed CWDM module provides transmission of all signals over a single optical fiber core for distances up to 40Km.

PSUs need to be ordered separately.

/ FEATURES

- Compact size – ideal for confined spaces, including camera poles and roadside cabinets
- -40°C to +75°C temperature maintains performance in extreme conditions
- Plug & Play – no need for any user configuration
- Dual SD/HD/3G-SDI video channels
- 4-Wire RS422 data channel
- 3 open collector alarm outputs for system status
- Integrated CWDM module provides full system operation over a single optical fiber core
- Supports distances up to 40Km
- Automatic cable equalization to ensure signal integrity
- Designed in the USA & UK. Manufactured in the United Kingdom
- AMG Lifetime Support Warranty

Specifications.

Video.

Channels	2
Format	SD, HD, 3G-SDI
Signal Type	SMPTE 259M, 292M & 424M
Data Rate	270Mbit/s to 2.97Gbit/s
Cable Equalization	Automatic Cable EQ (Belden 1694A cable) 250m @ 270Mbit/s, 140m @ 1.5Gbit/s, 80m @ 3Gbit/s

Data.

Data Protocol	RS422
Data Rate	0 to 200Kbps
Error Rate	< 10 ⁻⁹

Alarm Output.

Outputs	3
Type	Open Collector
Current Sink	115mA
Voltage	+60V _{DC} Maximum
Conditions	Power Failure Optical Link Failure Video Input Failure

Interfaces.

LED Indicators	Power Optical Link SDI Channel 1 SDI Channel 2
Power/Data/Alarms	1x 14 Position (7x2 Row) Harwin Connector M80-5401442
Video	2x Molex Coaxial Connectors 73404-2330 (75 Ohm)
Optical	1x LC Simplex Connector

Power.

Power Inputs	1
Operating Voltage	10-30V _{DC}
Power Consumption	4W Max
Protection	Reverse Polarity Overload Current

Optical.

Wavelength	1310nm/1490nm/ 1550nm/1610nm
Output Power	-5 to 0dBm per Wavelength
Fiber Type	9/125µm Singlemode
No. Fibers	1
Optical Connector	LC
Optical Budget	13dB Minimum
Rx Sensitivity	<-18dBm
Rx Saturation	>0dBm
Distance	0-40Km

Mechanical.

Housing	Anodised Aluminium
Dimensions:	(W x D x H) 43 × 190 × 96 mm 1.69 × 7.48 × 3.78 in
(Excluding Wall Brackets)	
IP Rating	IP40
Installation	Wall Mount or DIN-Rail
Weight:	0.70kg / 1.54lb

Environmental.

Operating Temp.	-40 to +75°C / -40 to +167°F
Storage Temp.	-40 to +85°C / -40 to +185°F
Humidity	5% to 95% (non-condensing)
MTBF	>250,000hours
MTBF Standard	Telcordia SR-332 GF 30°C
Heat Dissipation	14 BTU/h
Cooling	Passive Cooling
Noise Level	0 dBA

Regulatory.

Safety	IEC/EN 62368-1
EMI	EN 55032 Class A CISPR 32 FCC Part 15B Class A
EMS	EN 61000-4-2 (ESD) EN 61000-4-3 (RS) EN 61000-4-4 (EFT) EN 61000-4-5 (Surge) EN 61000-4-6 (CS) EN 61000-4-8 (PFMF)
Shock	IEC 60068-2-27
Free Fall	IEC 60068-2-32
Vibration	IEC 60068-2-6
Environmental	Reach, RoHS, WEEE
Supply Chain	NDAA & TAA Compliant

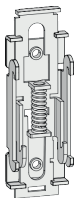
Part Numbers.

Industrial 2 Channel 3G-SDI, 1 Channel RS422 Modules

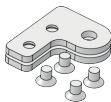
AMG5122TX	Dual Channel 3G-SDI, Single Channel RS422, Transmitter, Singlemode Fiber, Standalone
AMG5122RX	Dual Channel 3G-SDI, Single Channel RS422, Receiver, Singlemode Fiber, Standalone

Included Accessories.

DIN Rail Adapter
Wall Mounting Brackets (Type 1)
Wall Mounting Bracket (Type 2)



Rear Mounted Metal DIN Rail Clip & Screws For DIN Rail Mounting AMG5122 Products
2x Wall Mounting Brackets & Screws For Rear Wall / Surface Mounting AMG5122 Products
1x Wall Mounting Bracket & Screws For Side Wall / Surface Mounting AMG5122 Products



Recommended PSUs.

AMGPSU-W12-P25	Plug Top Mounting Lite Industrial Grade PSU, 0 to +70°C, 12VDC, 25W, UK/EU/US Plug Heads Included
AMGPSU-I12-P24	DIN-Rail Mounting Industrial Grade PSU, -40 to +70°C, 12VDC, 24W

In a continuing effort to improve and advance technology, product specifications are subject to change without notice. Please visit www.amgsystems.com for the latest product specifications.